HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN
B. Sc. PHYSICS :SEMESTER -VI

TYPE OF COURSE: MAJOR DISCIPLINE SPECIFIC COURSE (THEORY)

COURSE NAME: MATHEMATICAL PHYSICS, CLASSICAL - QUANTUM MECHANICS and MOLECULAR SPECTRA

PROGRAMME CODE: SCIUG103 COURSE CODE: SC23MJDSCPHY601

(Effective from June 2025 Under NEP–2020)

Total Credits: 04 Teaching Hoursper Week: 04 Teaching Hoursper Semester: 60	THEORY MAJOR	External Marks (SEE) - 50 Internal Marks (CCE) - 50

Course Objective:

- To provide various numerical techniques for bassel, spherical Bessel functions and can solve Leguerre, Legendre, Rodriguez's polynomials equations.
- To create awareness about motion of rigid body and get sufficient deep knowledge of Lagrange's and Hemilton's Equations and applications of them.
- To learn physical interpretations of the energy eigen functions, Properties of Stationary States, Coherent States
- To give physical interpretations of the energy eigen functions, Properties of Stationary States, Coherent States
- To learn characteristics of Types of Molecular Energy and Molecular Spectra
- To get sufficient knowledge of Rotational Spectra and Salient Features of Vibrational-Rotational Spectra.

Course Outcome: After the successful completion of the course students will be able to Understands,

- The various numerical techniques for bassel, spherical Bessel functions
- Legendre, Rodriguez's polynomials equations.
- about motion of rigid body and get sufficient deep knowledge of Lagrange's, Hemilton's Equations and applications of them.
- physical interpretations of the energy eigen functions, Properties of Stationary States, Coherent States
- physical interpretations of the energy eigen functions, Properties of Stationary States, Coherent States
- Characteristics of Molecular Energy and Molecular Spectra
- Rotational Spectra and Salient Features of Vibrational-Rotational Spectra and applications.

Syllabus

UnitNo.	Content	Credit	Lect. Hrs 60
Unit-1	**MATHEMATICAL PHYSICS:** **(a) Some Special Functions in Physics :** Legendre Differential Equation (6.1), Generating Function of Legendre Polynomial (6.2), Rodriguez's formula for Legendre Polynomial (6.3), Orthogonal Properties of Legendre Polynomial (6.4), Recurrence of relation for Legendre Polynomials (6.5), Bessel differential equation (6.6), Recurrence formulae Bessel's functions (6.7), Generating Function for *Jn(x)* (6.8), Orthogonality of Bessel's Functions (6.9), Hermite differential equation and Hermite Polynomial (6.11), Generating function of Hermite Polynomial (6.12), Recurrence formula for Hermite Polynomial (6.13), Rodriguez's formula for Hermite Polynomial (6.14), Orhogonality of the Hermite polynomials(6.15) *Related Examples, Problems & Short Questions* **Basic Reference:** Mathematical Physics by Satya Prakash, Pragati Prakashan (Reprint-2008)	1	15

Unit-II	**CLASSICAL MECHANICS :** **(a) Motion of Rigid Body :** Euler's theorem (10.1), angular momentum and kinetic energy (10.2), The inertia tensor (10.3), Euler's equations of motion (10.4) *Related Examples, Problems & Short Questions* **(b) Variational Principle: Lagrange's and Hemilton's Equations :** Configuration Space (11.1), Some Techniques of Caclulus of Variation (11.2), Applications of The Variational Principle (11.3), Hemilton's Principle (11.4). Equivalence of Lagrange's and Newton's Equations (11.5), Advantages of The Lagrangion Formulation-Electromechanical Analogies (11.6), Lagrange's undertermined Multipliers (11.7), Lagrange's Equation For Non- Holonomic System (11.8), Application of The Lagrangian Method of undetermined Multipliers (11.9), Hemilton's Equations of Motion (11.10), Some Applications of the Hamiltonian Formulation (11.11) *Related Examples, Problems & Short Questions* **Basic Reference**: Introduction to Classical Mechanics by Takawale and Puranik. McGraw Hill Education (India) private limited.	1	15
Unit-III	**QUANTUM MECHANICS :** **(a) Exactly Soluble Eigen Value Problems : The Simple Harmonic Oscillator** The Echrodinger Equation and Energy Eigenvalues (4.1), The Energy Eigen Functions (4.2), Properties of Stationary States (4.3), The Abstract Operator Method (4.4), Coherent States (4.5). **(b) Angular Momentum and Parity** The Angular momentum Operators (4.6),The Eigenvalue Equation for L 2 ; Separation of Variables (4.7), Admissibility Conditions on Solutions ; Eigenvalues (4.8), The Eigen Functions:Spherical Harmonics (4.9), Physical Interpretation (4.10), Parity (4.11), Angular Momentum in Stationary States of Systems with Spherical Symmetry (4.12), Solution of the Interior region (4.13) *Related Examples, Problems & Short Questions* **Basic Reference** : A text book of Quantum Mechanics by P.M. Mathews and K. Venkateshan, McGraw Hill education (India) private limited.2nd Edition	1	15
Unit-IV	**MOLECULAR SPECTRA :** **(a) Types of Molecular Energy and Molecular Spectra :** Separation of Electronic and Nuclear Motion: the Born Oppenheimer Approximation (17.1), Types of Molecular Spectra (17.2), Types of Molecular Energy States and Associated Spectra(17.3) **(b) Pure Rotational Spectra :** Salient Features of Rotational Spectra (18.1), Molecular Requirement for Rotational Spectra (18.2), Experimental Arrangement (18.3), The Molecule as a rigid Rotator: Explanation of Rotational Spectra (18.4), The Non-rigid Rotator (18.5), The Isotope Effect (18.6). **(c) Vibrational – Rotational Spectra :** Salient Features of Vibrational-Rotational Spectra (19.1), The Molecule as a Harmonic Oscillator (19.2), Anharmonic Oscillator (19.3), Vibrational Frequency and Force Constant for Anharmonic Oscillator (19.4), Isotope effect on Vibrational Levels(19.5) *Related Examples, Problems & Short Questions* **Basic Reference:** Atomic & Molecular-Spectra by RajKumar, KedarNath RamNath, Delhi.	1	15

: Further Reading – Other References:

1. Mathematical Physics by B.D.Gupta.
2. Mathematical Physics by H.K.Dass.
3. Mathematical Physics by P K Chattopadhyay (Second Edition), New age Int. publishers
4. Classical Mechanics, by Goldstein. Narosa Publishing House, NewDelhi.
5. Classical Mechanics by YasvantWaghmare.
6. Classical Mechanics by N. C. Rana and P. S .Joag,THM
7. Classical Mechanics by Gupta, Kumar and Sharma
8. Quantum Mechanics by Ghatak and Loknathan, The Macmillan Company of India Limited
9. Quantum Mechanics by Fschwabi, Narosa Publishing House, NewDelhi.
10. Quantum Mechanics by John, L. Powell and B.Crasemann.
11. Quantum Mechanics by Schiff
12. Molecular spectroscopy by Herz-Berg.
13. Molecular spectroscopy by Banewell
14. Atomic Physics by J B Rajam (7th Edition) S. Chand publication, Delhi
15. Elements of spectroscopy by S L Gupta, V Kumar, R C Sharma, (30th Edition) Pragati Prakashan

HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN

B. Sc. PHYSICS :SEMESTER -VI

TYPE OF COURSE: MAJOR DISCIPLINE SPECIFIC COURSE (THEORY)

COURSE NAME: STATISTICAL MECHANICS, SOLID STATE PHYSICS, ELECTRODYNAMICS AND ELECTRONICS

PROGRAMME CODE: SCIUG103 COURSE CODE: SC23MJDSCPHY601A

(Effective from June 2025 Under NEP–2020)

Total Credits: 04 Teaching Hoursper Week: 04 Teaching Hoursper Semester: 60	THEORY MAJOR	External Marks (SEE) - 50 Internal Marks (CCE) - 50

Course Objective:

- To provide concepts of a statistical mechanics and knowledge about partition functions.
- To create awareness about Gibbs Paradox.
- To Give basic knowledge superconductivity and application of it.
- To Understands the BCS theory.
- To study the electromagnetic and applications of potential.
- To provide an exposure about feedback amplifier and Transistor oscillators.

Course Outcome: After the successful completion of the course students will be able to,

- Understands concepts of a statistical mechanics and knowledge about partition functions and Gibbs paradox.
- Get sufficient basic knowledge superconductivity and application of it. Also known about BCS theory.
- Solve certain special physical problems regarding mathematical aspects.
- Understands concept of the electromagnetic and applications of potential, also aware about Laplace's Equation, Boundary conditions, Cartesian Coordinates, Spherical coordinates
- Get sufficient knowledge of Feedback Amplifier and Transistor Oscillators (Sinusoidal).Also Create awareness about Principle of Feedback Amplifiers and Advantages of Negative Feedback.
- Get an exposure to various type of oscillators.

Syllabus

UnitNo.	Content	Credit	Lect. Hrs 60
Unit-1	**STATISTICAL MECHANICS :** **(a) Some Application of Statistical Mechanics:** Rotating Bodies (6.1), The Probability Distribution for Angular Momenta and Angular Velocities of Rotation of molecules (6.2), Thermodynamics (6.3), Reversible and Irreversible processes (6.3.1), The Laws of Thermodynamics (6.3.2) ((i) Zeros (ii) First Law (iii) Second Law (iv) Third Law), Statistical interpretation of the basics thermodynamic variables (6.4) & (6.4.1 to 6.4.8), Physical Interpretation of **α** (6.5), Chemical Potential in the equilibrium state (6.6), Thermodynamic functions in terms of grand partition function (6.7), Ideal gas (6.8), Gibbs's Paradox (Inclusive Sackur-Tetrode equation) (6.9), The Equipartition Theorem (6.10), The Statistics of Para magenetism (6.11), Potential Energy of a Magnetic Dipole in a Magnetic Field (6.11.1), Curle's Law (6.11.2), Thermal Disorder in a Crystal Lattice (6.12), Non-Ideal Gases (6.13). *Related Examples, Problems & Short Questions* **Basic Reference :**Fundamentals of Statistical Mechanics by B. B. Laud. (Second Edition) New Age International Publisher	1	15

Unit-II	**SOLID STATE PHYSICS:** **Superconductivity :** Phenomena without observable Quantization(15.1), Zero resistance and persistent currents(15.1.1), Perfect Diamagnetisms : Meissner Effect(15.1.2), London Equation (15.1.3), Critical Field : Type I and Type II super conductors (15.1.4), Thermodynamic properties (15.1.5), Energy gap (15.2), Properties Dependent on Energy gap (15.3), Heat Capacity (15.3.1), Thermal Conductivity (15.3.2), Absorption of Electromagnetic Radiation (15.3.3), Normal Tunnelling (15.3.4), Isotope Effect (15.4), BCS Theory : A qualitative approach (15.5), Cooper pair formation (15.5.1), BCS ground state (15.5.2), Important predictions of the BCS theory and comparison with experiments (15.6), Critical temperature (15.6.1), Ginzburg-Landau Theory (15.7), Magnetic flux Quantization (15.7.1), Coherence Length (15.7.2), Type-II superconductivity (15.7.3), Josephson tunneling (15.7.4), Applications(15.9). *Related Examples, Problems & Short Questions* **Basic Reference:** Elements of Solid State Physics by J.P. Srivastava, PHI New Delhi 2006 (2nd Edition)	1	15
Unit-III	**ELECTRODYNAMICS** **ELECTROMAGNETICS (a) Potentials :** Laplace's Equation (3.1), Introduction (3.1.1), Laplace's Equation in one dimensions (3.1.2), Laplace's Equation in two dimensions(3.1.3), Laplace's Equation in three dimensions (3.1.4), Boundary conditions and Uniqueness theorems (3.1.5), Conductors and the second Uniqueness theorem (3.1.6), The method of images (3.2), The classic image problem (3.2.1), Induced surface charge (3.2.2), Force and energy (3.2.3), other image problems (3.2.4) Separation of variables (3.3), Cartesian Coordinates (3.3.1), Spherical coordinates (3.3.2), Multipole Expansion (3.4), Approximate Potential at large distances (3.4.1), The monopole and dipole terms (3.4.2), Origin of Coordinates in multipole Expansions (3.4.3). *Related Examples, Problems & Short Questions* **Basic Reference:** Introduction to Electrodynamics by David J. Griffiths. 4th Edition Cambridge University press.	1	15
Unit-IV	**ELECTRONICS:** **(a) Feedback Amplifier :** Feedback (8.1), Principle of Feedback Amplifiers (8.2), Advantages of Negative Feedback (8.3), Reasons for Negative Feedback (8.4 a,b,c,d), Negative feedback circuits (8.5). **(b) Transistor Oscillators (Sinusoidal):** Introduction to oscillators : Types, components and Barkhausen criterion, Tuned Collector Oscillators (11.1), Tuned Emitter Oscillator (11.3), Tuned Base Oscillator (11.3), Hartley Oscillator (11.4), Colpitt's Oscillators (Circuit operation and alternative treatment only)(11.5), CLAPP Oscillator(11.6), Phase Shift oscillators (11.7), R-C-Oscillator (11.7.1), Wien Bridge Oscillator (11.7.2), Crystal Oscillator (11.8). *Related Examples, Problems & Short Questions* **Basic Reference:** Hand Book of Electronics by Gupta and Kumar. (46th revised Edition 2019)	1	15

	: Further Reading – Other References: 1. Statistical Mechanics and Properties of Matter by E.S.R.Gopa 2. Statistical Mechanics by B K Agarwal& Melvin Eisner, Wiley Eastern 3. Statistical Mechanics - Theory and Application by S K Sinha, TMHPubCo.Ltd. New Delhi 4. Statistical Mechanics by R K Pathariya, Pegramon Press 5. Solid State Physics by C. Kittel. John Willy and Sons. 6. Solid State Physics by Saxena. PragatiPrakashan. 7. Solid State Physics by C. M.Kachhawa. 8. Electromagnetics by B. B. Laud. Willley Eastern	